

CHEMICAL CHANGES – THE REACTIVITY SERIES

KEY VOCABULARY

Reactivity series: metals listed from most to least reactive.

Displacement: a more reactive metal pushes out a less reactive one.

Oxidation: loss of electrons (metal forming an ion).

Native metal: an unreactive metal found as the element, e.g. gold.

COMMON MISCONCEPTIONS

~~— Gold is reactive because it is valuable.~~

✓ Gold is very unreactive — that's why it stays shiny.

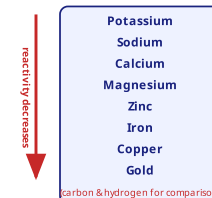
~~— All metals react with water.~~

✓ Only reactive metals react with cold water.

~~— A less reactive metal displaces a more reactive one.~~

✓ Only a MORE reactive metal can displace another.

REACTIVITY SERIES



1 ORDER METALS

Put these in order of reactivity (most first): copper, potassium, zinc. [1 mark]

2 DISPLACEMENT

Predict what happens when iron is added to copper sulfate solution. [2 marks]

3 EXPLAIN REACTIVITY

Explain what makes one metal more reactive than another. [2 marks]

Think about losing electrons to form ions.

4 REACTION WITH ACID

Explain why magnesium fizzes faster than zinc in acid. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A ____ reactive metal displaces a ____ reactive metal from its compound.

more

less

lose

gain

6 WRITE AN EQUATION

Write the equation for zinc displacing copper from CuSO_4 . [2 marks]